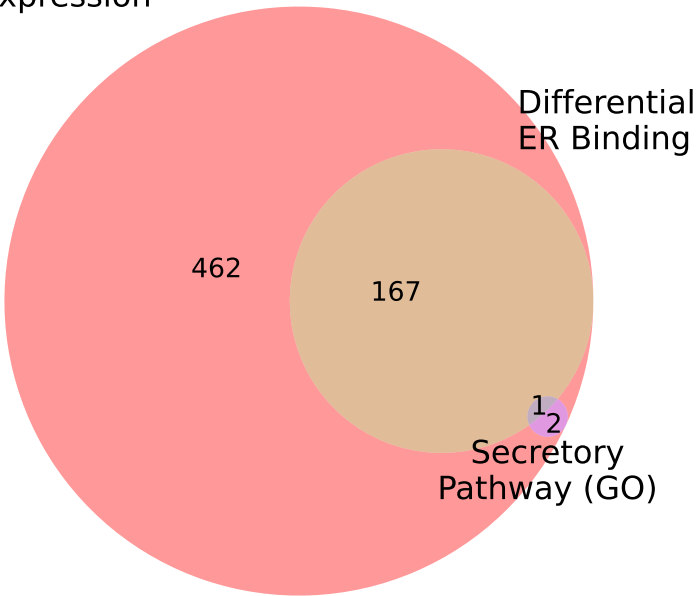
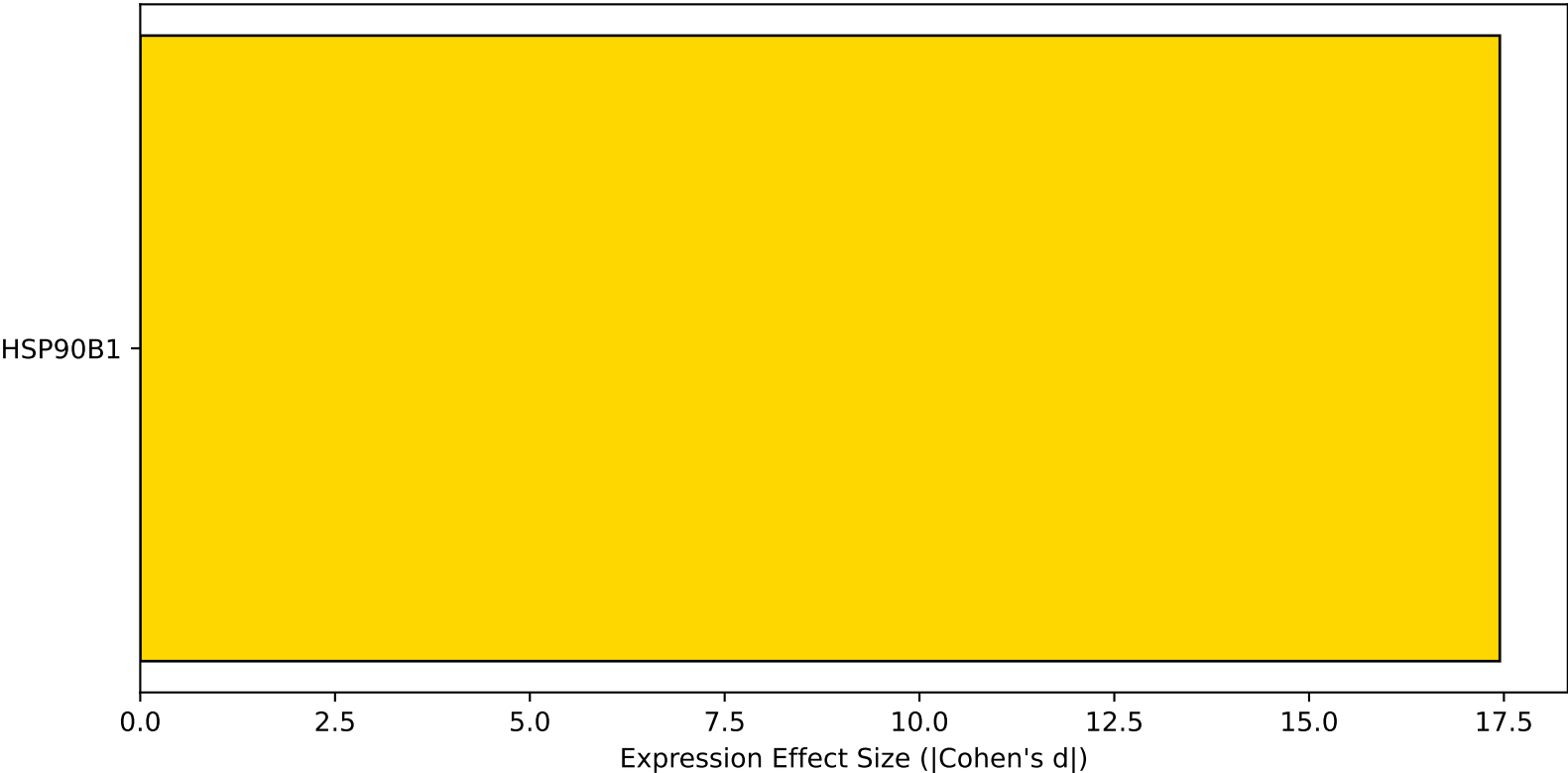


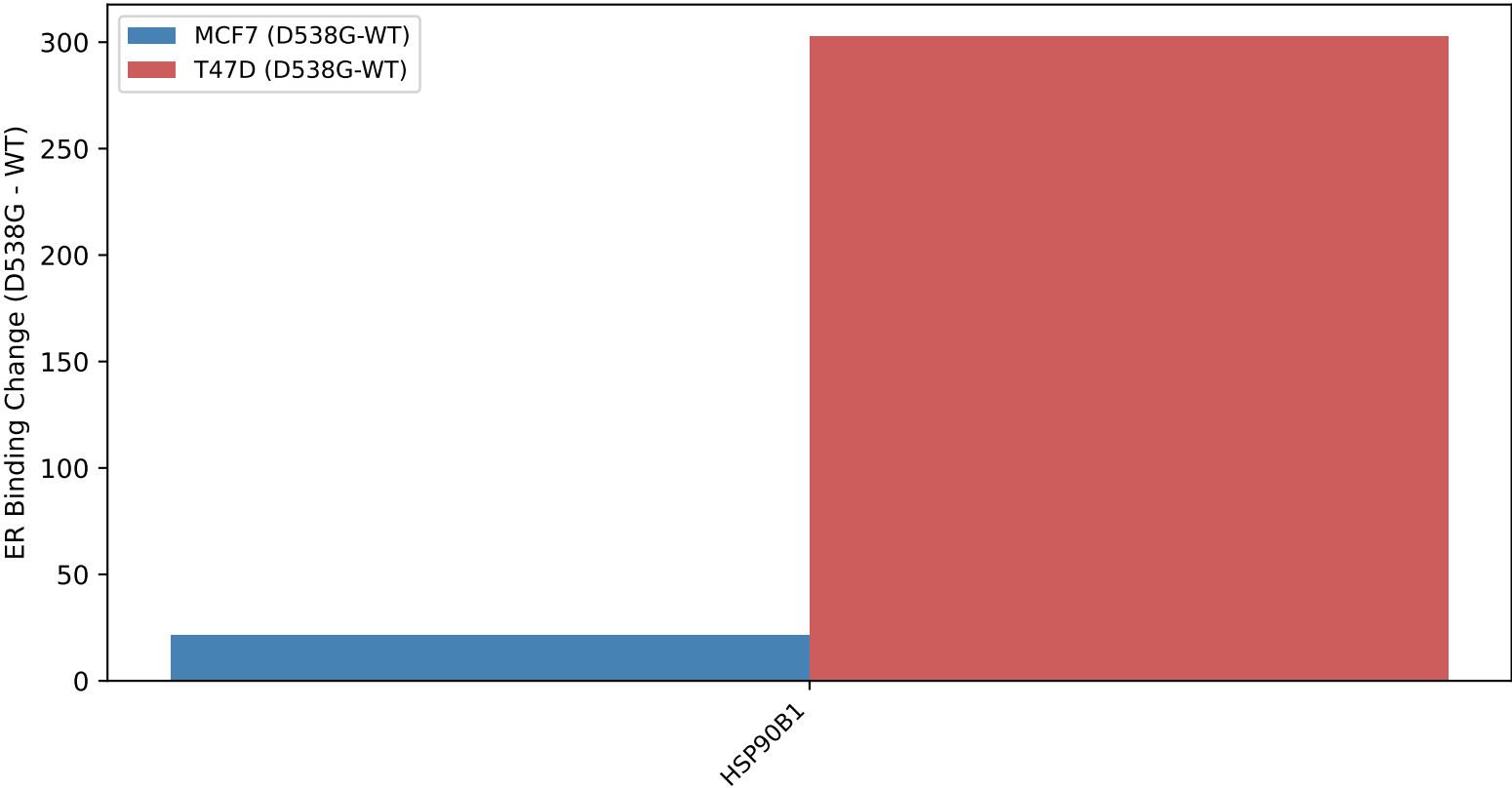
A. Three-Way Venn: Winnowing Candidates
(GSE89888 + GSE125117 + Gene Ontology)
Opposite
Expression



B. ALL 1 Genes in Triple Intersection
(Gold = HSP90B1)



C. ER Binding Changes for Triple Intersection Genes
(GSE125117)



THREE-WAY VENN: WINNOWING TO HSP90B1

FILTER 1: Opposite Expression (GSE89888)

- 632 genes with opposite D538G effect (MCF7 vs T47D)

FILTER 2: Differential ER Binding (GSE125117)

- 168 genes with ER binding change at promoter upon D538G mutation

FILTER 3: Secretory Pathway (Gene Ontology)

- 3 genes in ER/secretory pathway (GO terms for protein folding, ERAD, transport)

TRIPLE INTERSECTION:

- 1 genes meet ALL THREE criteria

HSP90B1 STATUS:

- In intersection: YES
- Rank: #1/1

CONCLUSION:

Starting from 632 genes, the triple intersection identifies only 1 candidates that are:

1. Oppositely regulated by D538G
2. Have differential ER binding
3. Are in the secretory pathway